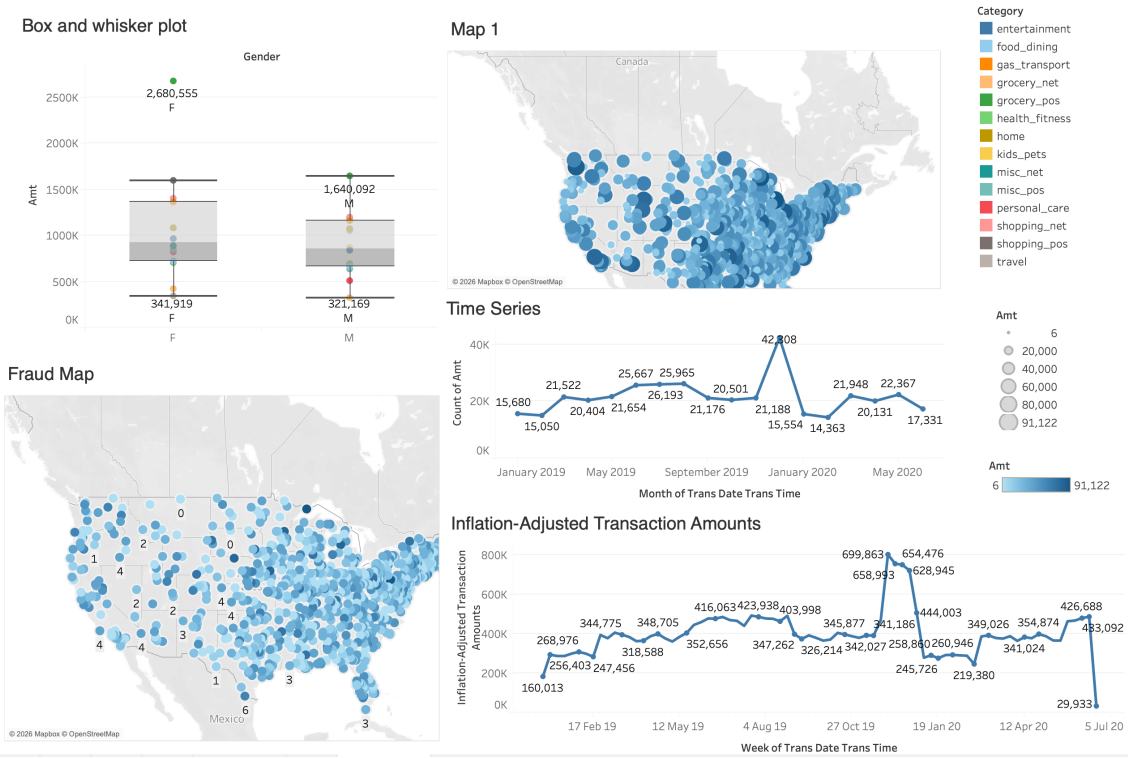
# Tableau Dashboard Insights

The Tableau dashboard was developed to provide an interactive view of transaction behaviour, fraud distribution, geographical spread, and time-based spending patterns. The visualizations helped summarize the key trends in the dataset and supported a better understanding of fraud-related behaviour.




## Dashboard Summary Insights:

- The **box and whisker plot** shows that transaction amounts vary across both male and female customers, with significant outliers present in each group. Female customers appear to have the highest extreme transaction amount, indicating the presence of unusually large transactions that may require further investigation.
- The **transaction map** highlights that transactions are widely distributed across the United States, with denser activity visible in the eastern and southeastern regions. This suggests that transaction activity is concentrated in certain population-heavy areas.
- The **fraud map** indicates that fraudulent transactions are geographically dispersed rather than limited to a single region. This suggests that fraud is a broad behavioural issue and not only a location-specific problem.
- The **time series chart** shows fluctuations in transaction counts over time. Transaction activity generally remains stable, but there is a noticeable spike around the beginning of 2020, which may indicate seasonal behaviour, reporting variation, or a temporary surge in activity.

- The **inflation-adjusted transaction amount trend** reveals strong variation over time, with several noticeable peaks and drops. This indicates that transaction value is not constant and changes significantly across time periods.
- The dashboard as a whole suggests that **high-value transactions, temporal spikes, and unusual observations across gender and geography** are important areas for fraud monitoring.
- The visualizations improve decision-making by allowing stakeholders to quickly identify trends, outliers, and suspicious behaviour patterns in the transaction data.

## Conclusion

The Tableau dashboard effectively summarizes the major patterns in the fraud detection dataset. It shows that transaction behaviour varies across customer groups, locations, and time, while also highlighting abnormal transaction amounts and broad geographic fraud presence. These insights are valuable for monitoring fraud risk and supporting further predictive analysis.